Toward a four-dimensional Yang–Mills construction

Notebook 18: integrating overlapping cutoff interactions

Research working document

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Abstract

We extend the Gaussian cutoff expansion to interactions whose supports overlap and can have arbitrarily many cells. An explicit weighted local norm controls their expansion. The covariance forest is applied after grouping overlapping factors, but its derivatives retain their original cell endpoints. This prevents uncontrolled powers of the grouped support size at high-degree vertices. We obtain a nonvanishing partition function, a holomorphic logarithm with summable activities, and a bound linear in the interaction norm for its difference from the pure Gaussian cutoff logarithm. A support-algebra construction also gives strictly local background activities when the precision has a small local decomposition. For a real baseline, a tunable relative Gaussian comparison reduces the density cost per cell from a fixed exponential in its dimension to an exponential with an arbitrarily small coefficient. The theorems specify the covariance and coefficient estimates that the Yang–Mills quantum step must supply, and a final section identifies the corresponding original Bałaban integration route.

1 The integration problem addressed here

The nonlinear Wilson action and its coordinate factors generate interactions supported on several overlapping cells. Treating each connected collection as a single large Gaussian cell can introduce a loss growing with both its support size and its branching degree. The proof below keeps the original cell structure through the covariance derivatives and sums the resulting degree factorials exactly. The original Gaussian cell dimension stays fixed.

This is an abstract integration theorem with explicit hypotheses. The homogeneous flat pivot Gaussian has the spectral and locality estimates from Notebooks 6 and 7; Notebook 17 records its joint fluctuation bounds. To apply the present theorem to the complete Yang–Mills integrand, the action, Haar and Jacobian contributions must be represented with the stated weighted local coefficient norm on the actual cutoff domain.

The overlap, background-support, tunable-density and periodic arguments below have passed separate internal mathematical audits.

2 Overlapping interactions in a Gaussian cutoff integral

The disjoint-cell cutoff expansion extends to overlapping interaction supports, including supports of unbounded size, under an explicit weighted multiplier norm. The essential step is to retain the original cell endpoints when differentiating covariance edges between larger interaction molecules. This controls the degree factorials without enlarging the fixed Gaussian cell dimension. The statement below is abstract; it does not verify the fluctuation coordinates or interaction bounds of the Yang–Mills construction.

2.1 Assumptions and quantitative statement

Let Λ be a finite metric set of cells, each carrying $x_i \in \mathbb{R}^D$, with D fixed. Let A vary in an open complex background domain. The complex symmetric covariance $C(A)$ is holomorphic and obeys, uniformly in volume and background,

$$\operatorname{Herm} C \geq \lambda I, \qquad \|C\| \leq \Lambda_0, \qquad 0 < \lambda \leq \Lambda_0. \tag{1}$$

For $i \neq j$ assume

$$\|C_{ij}\| \leq J_{ij} = J_{ji}, \qquad J_{ii} = 0, \qquad A_\kappa = \sup_i \sum_j e^{\kappa d(i,j)} J_{ij} < \infty. \tag{2}$$

Write $\ell(S)$ for the minimum metric spanning-tree length of a nonempty set S, with $\ell(\{i\}) = 0$. Gaussian expectations use

$$p_C(x) = (2\pi)^{-D|\Lambda|/2} (\det C)^{-1/2} \exp(-x^T C^{-1} x/2),$$

with the analytic determinant branch on the accretive cone.

Fix the real cutoff

$$\chi_i = \mathbf{1}_{\{\|x_i\|_2 \leq R\}}, \qquad \chi_Y = \prod_{i \in Y} \chi_i.$$

Consider a finite labelled family of nonempty interaction supports $Y \subset \Lambda$. Distinct labels may share a support. The functions $V_Y(A, x_Y)$ are measurable in the real variables and holomorphic in A where $\chi_Y = 1$, and

$$v_Y = \sup_{A,\, \chi_Y = 1} |e^{V_Y(A, x_Y)} - 1| < \infty. \tag{3}$$

No values outside that cutoff region or real-field derivatives are required. Define $w_Y = \chi_Y(e^{V_Y} - 1)$, extended by zero. Set

$$\delta = \lambda/2, \qquad V_* = 8\Lambda_0^2/\lambda, \qquad K_G = (2V_*/\lambda)^{D/2},$$

$$p_R = 2^{D/2} e^{-R^2/(4V_*)}, \qquad \tau = K_G p_R, \qquad c_G = e^2/\delta, \tag{4}$$

$$\epsilon = \sup_i \sum_{Y \ni i} v_Y K_G^{|Y|} e^{4|Y| + \kappa \ell(Y)}, \qquad \eta = e^3 \tau + \epsilon, \qquad \eta_* = \frac{1}{4 \max(1, c_G A_\kappa)}. \tag{5}$$

The interaction sum counts labels. These sufficient constants are not asserted to be optimal.

Theorem 2.1 (Overlapping Gaussian cutoff interactions)**.** *If $\eta \leq \eta_*$, then*

$$Z_V(A) = \mathbb{E}_{C(A)} \left[\prod_{i \in \Lambda} \chi_i \exp\left(\sum_Y V_Y \right) \right]$$

is nonzero and has a holomorphic logarithm defined by a convergent expansion

$$\log Z_V(A) = \sum_{\varnothing \neq X \subset \Lambda} U_V(X; A), \qquad \sup_i \sum_{X \ni i} e^{\kappa \ell(X) + |X|} \|U_V(X; \cdot)\|_\infty \leq 2\eta. \tag{6}$$

The constants do not depend on the number or sizes of supports. If

$$e^3 \tau \leq \eta_*/4, \qquad \epsilon \leq \eta_*/4, \tag{7}$$

the canonical expansion also satisfies

$$\sup_i \sum_{X \ni i} e^{\kappa \ell(X) + |X|} \|U_V(X; \cdot) - U_0(X; \cdot)\|_\infty \leq 8\epsilon, \tag{8}$$

where U_0 is the pure Gaussian cutoff forest/Mayer expansion. The support label X concerns fluctuation cells; strict locality in the background is a separate assertion.

2.2 First expansion: disjoint interaction molecules

Put $q_i = -\mathbf{1}_{\mathrm{bad},i}$. The identity

$$\prod_i \chi_i e^{\sum_Y V_Y} = \prod_i (1+q_i) \prod_Y (1+w_Y) \tag{9}$$

is exact, and $q_i w_Y = 0$ whenever $i \in Y$. For a nonempty set S let

$$a_S(x_S) = \sum_{\substack{\mathcal{F}\neq\varnothing,\ \cup_{Y\in\mathcal{F}} Y = S \\ \mathcal{F} \text{ overlap-connected}}} \prod_{Y\in\mathcal{F}} w_Y(x_Y),$$

$$M_S = \sum_{\substack{\mathcal{F}\neq\varnothing,\ \cup_{Y\in\mathcal{F}} Y = S \\ \mathcal{F} \text{ overlap-connected}}} \prod_{Y\in\mathcal{F}} v_Y. \tag{10}$$

Each label is used at most once in these sums. Connectivity refers to the intersection graph of the chosen supports. Then $|a_S| \le M_S$. Define $f_S = a_S$ for nonsingletons and $f_{\{i\}} = a_{\{i\}} + q_i$. Grouping overlap components in (9) gives exactly

$$\prod_i \chi_i e^{\sum_Y V_Y} = \sum_{\mathcal{P} \text{ pairwise disjoint}} \prod_{S\in\mathcal{P}} f_S(x_S). \tag{11}$$

The grouping is unique: interaction components and bad singleton marks have disjoint supports. Introduce

$$b(S) = K_G^{|S|} M_S + \tau \mathbf{1}_{\{|S|=1\}}. \tag{12}$$

Lemma 2.2 (Overlap-animal estimate)**.** *If $\epsilon \le 1$, then*

$$\sup_i \sum_{S\ni i} b(S) e^{\kappa\ell(S)+3|S|} \le e^3\tau + \epsilon = \eta. \tag{13}$$

Proof. Set $u_Y = v_Y K_G^{|Y|} e^{3|Y|+\kappa\ell(Y)}$. For an overlap-connected family, $\ell(\cup Y) \le \sum_Y \ell(Y)$ and $|\cup Y| \le \sum_Y |Y|$: join spanning trees of the supports, which share vertices along their intersection graph. Since $K_G \ge 1$, its contribution to the interaction part of (13) is at most $\prod_Y u_Y$.

Majorize a connected family anchored at i by rooted spanning trees of its intersection graph, choosing a root support containing i. Dropping distinctness of labels only increases the sum. The nonnegative rooted-tree majorant is the monotone iterative solution of

$$T_Y = u_Y \exp\left(\sum_{Z:\, Z\cap Y\neq\varnothing} T_Z \right).$$

If $T_Z \le u_Z e^{|Z|}$, then

$$\sum_{Z:\, Z\cap Y\neq\varnothing} T_Z \le \sum_{j\in Y} \sum_{Z\ni j} u_Z e^{|Z|} \le \epsilon|Y| \le |Y|.$$

Thus the bound persists through iteration, and $\sum_{Y\ni i} T_Y \le \epsilon$. Every original connected family is included at least once. The remaining bad singleton term is $e^3\tau$. □

2.3 Covariance forests with original cell endpoints

Fix disjoint primitive supports $S_1, \dots, S_n$. Introduce a covariance parameter s_{ab} for each pair of molecules, multiplying all blocks between S_a and S_b and leaving internal blocks unchanged. Partition masks in these parameters are direct sums of actual principal submatrices of C. Their convex hull

retains (1), so the BKAR forest formula applies there. At a forest mask both the covariance and the product of primitive functions factorize over the forest components.

For a molecule-tree T, expand each covariance derivative into its original endpoint cells:

$$\partial_{s_{ab}} = \sum_{i\in S_a,\, j\in S_b} \partial_{z_{ij}}, \tag{14}$$

where z_{ij} multiplies just C_{ij} and C_{ji}. Choose one endpoint pair for each tree edge and let d_i count the resulting degree of cell i. No chosen endpoint pair repeats. At the real forest mask, allow complex increments with radii

$$r_{ij} = \frac{\delta}{\max(d_i, d_j) J_{ij}}. \tag{15}$$

Zero covariance edges contribute zero and can be omitted. Every block row and column of the perturbation has sum at most δ, hence its operator norm is at most δ. On the whole polydisc,

$$\operatorname{Herm} M \geq \lambda I/2, \quad \|M\| \leq 2\Lambda_0, \quad \operatorname{Herm} M^{-1} \geq V_*^{-1} I, \quad |\det M| \geq (\lambda/2)^{D|\cup S_a|}.$$

Thus $|p_M|$ is bounded by $K_G^{|\cup S_a|}$ times the product real Gaussian density with cell covariance $V_* I$. For singletons use $|f_i| \leq M_i + \mathbf{1}_{\mathrm{bad},i}$ and its independent Gaussian tail bound p_R. For other supports use $|f_S| \leq M_S$. This proves

$$\left| \mathbb{E}_M \prod_a f_{S_a} \right| \leq \prod_a b(S_a).$$

Cauchy's estimate, once in every chosen covariance edge, yields

$$\left| \partial_{T,\ \text{endpoints}} \mathbb{E}_{C(s)} \prod_a f_{S_a} \right| \leq c_G^{n-1} \prod_a b(S_a) \prod_i d_i! \prod_{ij\in T} J_{ij}. \tag{16}$$

Indeed, $\prod_{ij} \max(d_i, d_j) \leq \prod_i d_i^{d_i} \leq e^{2(n-1)} \prod_i d_i!$. The factorials are attached to original cells, including degree-zero cells with $0! = 1$, and not to the sizes or degrees of whole molecules. Neither indicators nor interaction multipliers are differentiated.

Forest components now regroup (11) into the exact hard-core gas

$$Z_V = \sum_{\mathcal{Q}\ \text{pairwise disjoint}} \prod_{X\in\mathcal{Q}} \zeta(X). \tag{17}$$

Here $\zeta(X)$ sums partitions of X into disjoint primitive supports and molecule trees with their BKAR integrals. The one-primitive case is its un-interpolated expectation.

2.4 Endpoint factorials and the rooted-tree sum

Lemma 2.3. *Under (13) and $\eta \leq \eta_*$,*

$$\sup_i \sum_{X\ni i} |\zeta(X)| e^{\kappa \ell(X) + 2|X|} \leq \eta. \tag{18}$$

Proof. Put $\widehat{b}(S) = b(S) e^{\kappa\ell(S)+2|S|}$ and $\widehat{J}_{ij} = e^{\kappa d(i,j)} J_{ij}$. The union weight is dominated by these support weights and endpoint edge weights: spanning trees inside the supports together with the chosen covariance edges connect their union.

Root a molecule tree and retain the incoming cell at each nonroot vertex. Let F_i majorize planted trees whose root molecule contains i and carries one extra incoming half-edge at i. This half-edge

has no covariance weight; it increments the cell degree. Allowing repeated or overlapping primitive supports enlarges the positive majorant. With $t_k = c_G \sum_j \widehat{J}_{kj} F_j$, its recursion is

$$F_i = \sum_{S \ni i} \widehat{b}(S)(1-t_i)^{-1} \prod_{k \in S} (1-t_k)^{-1}. \tag{19}$$

To verify its normalization, suppose n_k children depart from cell k. The unordered-child factor, after summing endpoint assignments, is $1/\prod_k n_k!$. The cell-degree factorial in (16) is $\prod_k n_k!$, except that the incoming cell has $(n_i+1)!$. The residual factors are 1 at every ordinary cell and n_i+1 at the incoming cell. Their nonnegative geometric series give exactly (19). This can equivalently be obtained from ordered children weighted by the inverse factorial of their total number. It is the rooted labelled-tree exponential series; repeated labels are used only for the upper bound.

Define the recursion by finite-height truncation. If $F_j \le 2\eta$, then $t_k \le 2c_G A_\kappa \eta \le 1/2$, and

$$F_i \le 2 \sum_{S \ni i} \widehat{b}(S) 2^{|S|} \le 2 \sum_{S \ni i} \widehat{b}(S) e^{|S|} \le 2\eta.$$

This gives a convergent planted majorant. At an unplanted root the extra factor $(1-t_i)^{-1}$ is absent, so its sum over root supports containing i is at most $\sum_{S \ni i} \widehat{b}(S) 2^{|S|} \le \eta$. In an original disjoint primitive family whose union contains i, precisely one support contains i; root the tree there. This proves (18).

In particular a molecule of m cells with equal endpoint parameter t has unplanted generating function $(1-t)^{-m}$. Its degree-d coefficient is $(m)_d/d!$, a rising factorial divided by $d!$. Replacing this by the crude block bound $m^d d!$ would lose the support-exponential control. □

2.5 Logarithm and analytic comparison

For $z'(X) = |\zeta(X)| e^{\kappa \ell(X) + |X|}$, (18) gives the Kotecky–Preiss condition with size function $|X|$:

$$\sum_{Y \colon Y \cap X \neq \varnothing} z'(Y) e^{|Y|} \le |X| \eta \le |X|.$$

The absolutely convergent Mayer logarithm, grouped by the union of each connected overlap cluster, defines U_V. The union weight is at most the product of its polymer weights. Anchor a cluster at i by distinguishing a polymer containing i and apply the pinned cluster estimate. It gives

$$\sup_i \sum_{X \ni i} e^{\kappa \ell(X) + |X|} |U_V(X)| \le \sup_i \sum_{Y \ni i} |\zeta(Y)| e^{\kappa \ell(Y) + 2|Y|} \le \eta. \tag{20}$$

Uniform background suprema can be taken at every preceding step. Locally uniform convergence proves holomorphy and identifies Z_V with the exponential of this logarithm, proving nonvanishing and (6).

For (8), replace every w_Y by $t w_Y$ while retaining the bad singleton marks. This interpolates bounded multipliers. Its interaction norm is $|t|\epsilon$. If $\epsilon > 0$, set $\rho = \eta_*/(2\epsilon) \ge 2$. Under (7), all estimates hold for $|t| \le \rho$ because $\eta(t) \le 3\eta_*/4$. The canonical activity vector is holomorphic on this disk in the rooted activity norm and bounded by $2\eta_*$. Banach-valued Cauchy therefore gives

$$\|U(1) - U(0)\| \le \frac{2\eta_*}{\rho - 1} \le 8\epsilon.$$

The case $\epsilon = 0$ is immediate. At $t = 0$ only bad singleton primitives remain, so $U(0)$ is the original pure Gaussian cutoff forest/Mayer expansion. This completes the proof.

2.6 Scope of the abstract result

The same conclusion holds for countably many labelled interactions under (5), defining the integrand by the uniformly convergent product of their bounded multipliers on the cutoff, or equivalently by its finite-family limit. No convergence of the chosen logarithms V_Y themselves is asserted. If $u_Y = \sup_{\text{cutoff}} |V_Y| \le 1$, then $v_Y \le 2u_Y$ is a sufficient conversion to the required norm. Finite range is optional; the weighted summability is the actual hypothesis. The factor $K_G^{|Y|}$ budgets the Gaussian density cost but does not allow the original cell dimension D to grow with the regulator.

The argument establishes locality of support labels in fluctuation space. A covariance depending on a remote background variable may give every activity the same remote dependence, even if the covariance is diagonal. Strict background support consequently requires the separate local-precision/support-algebra hypotheses. Actual Yang–Mills use also requires the stated independent cell coordinates, uniform complex spectrum, fixed cutoff domain and weighted interaction bounds. None is inferred here from a global interaction stability estimate or from pointwise inverse bounds.

The imported expansion results are the BKAR forest formula on the convex hull of partition masks (Abdesselam–Rivasseau, arXiv:hep-th/9409094) and the Kotecky–Preiss hard-core polymer theorem, *Communications in Mathematical Physics* **103** (1986), 491–498. The molecule grouping, cell-endpoint Cauchy estimate and endpoint-factorial tree sum above supply the new overlapping-support step.

3 Strict background support for overlapping interactions

The preceding expansion can also be made strictly local in its background variables. This requires a background-independent baseline and a small local precision decomposition, in addition to local background dependence of the interactions. The next statement checks these additional inputs explicitly.

3.1 Baseline, local precision and constants

Let Λ be a finite subset of $\mathbb{Z}^d$ with the ambient lattice graph metric and fixed cell dimension D. The background domain is a product of complex coordinate domains. Write

$$P(A) = P_0 + \sum_Y B_Y(A|_Y), \qquad C_0 = P_0^{-1}, \tag{21}$$

where C_0 is complex symmetric and independent of every background variable, each B_Y is complex symmetric, has rows and columns supported in Y, and reads only $A|_Y$. Assume

$$\operatorname{Herm} C_0 \ge \lambda I, \qquad \|C_0\| \le \Lambda_0, \qquad 0 < \lambda \le \Lambda_0, \tag{22}$$

and, for fixed $0 < \kappa < m$,

$$\begin{aligned} K &= \sup_b \sum_c e^{(\kappa+2m)d(b,c)} \|(C_0)_{bc}\| < \infty, \\ \nu &= \sup_b \sum_{Y \ni b} \sum_{c \in Y} e^{(\kappa+2m)\ell(Y)} \|(B_Y)_{bc}\|_\infty < \infty. \end{aligned} \tag{23}$$

Symmetry gives the same column bounds. All constants are uniform in volume and background. Define

$$\begin{aligned} &\delta = \lambda/4, && V_* = 8\Lambda_0^2/\lambda, && J_{bc} = Ke^{-md(b,c)} \quad (b \ne c), \\ &A_\kappa = K \sum_{z \ne 0} e^{-(m-\kappa)|z|}, && \bar{K} = K + \delta, && a = \lambda/2, \\ &s = \min\{m/2,\ aem/(4\bar{K})\}, && L = \frac{4}{\lambda} \sum_{z \in \mathbb{Z}^d} e^{-s|z|}. \end{aligned} \tag{24}$$

Set $J_{bb} = 0$. Require $K\nu < 1$ and put

$$v = \frac{K^2\nu}{1 - K\nu}, \qquad v_1 = v(1 + \delta/K), \qquad u = Lv_1, \qquad w = \frac{L^2 v_1}{1 - u}. \tag{25}$$

The sufficient precision smallness conditions are

$$v_1 \le \lambda/4, \qquad u \le 1/2, \qquad w \le 1/(2V_*). \tag{26}$$

They are chosen before increasing the cutoff radius. The covariance is $C(A) = P(A)^{-1}$, given by its convergent Neumann series.

Retain the fixed cutoff and interaction multiplier hypotheses of (3), now requiring each V_Y to read background only in Y. Define

$$\begin{aligned} &\widehat{K}_G = (8V_*/\lambda)^{D/2}, && \widehat{p}_R = 2^{D/2} e^{-R^2/(8V_*)}, && \widehat{\tau} = \widehat{K}_G \widehat{p}_R, \\ &\widehat{\epsilon} = \sup_i \sum_{Y \ni i} v_Y \widehat{K}_G^{|Y|} e^{4|Y| + \kappa \ell(Y)}, && \widehat{\eta} = e^3 \widehat{\tau} + \widehat{\epsilon}, && \widehat{\eta}_* = \frac{1}{4 \max(1, e^2 A_\kappa/\delta)}. \end{aligned} \tag{27}$$

Theorem 3.1 (Strictly local overlapping-interaction activities)**.** *Under (21)–(26) and $\widehat{\eta} \le \widehat{\eta}_*$, the logarithm has a convergent representation by connected lattice supports:*

$$\log Z_V(A) = \sum_H E_{\Lambda,H}(A|_H), \qquad \sup_b \sum_{H \ni b} e^{a_*|H|} \|E_{\Lambda,H}\|_\infty \le 2 S_{\kappa/2} \widehat{\eta}, \tag{28}$$

where

$$a_* = \min(1/2, \kappa/4), \qquad S_{\kappa/2} = \sum_{z \in \mathbb{Z}^d} e^{-(\kappa/2)|z|}.$$

Each activity is holomorphic and reads only the indicated background coordinates. Supports may lie in the lattice hull of Λ. If

$$e^3 \widehat{\tau} \le \widehat{\eta}_*/4, \qquad \widehat{\epsilon} \le \widehat{\eta}_*/4, \tag{29}$$

the same canonical choice obeys

$$\sup_b \sum_{H \ni b} e^{a_*|H|} \|E_{\Lambda,H,V} - E_{\Lambda,H,0}\|_\infty \le 8 S_{\kappa/2} \widehat{\epsilon}. \tag{30}$$

3.2 The support algebra and the actual marginal

For a nonempty moment support M, let $\ell_M(X)$ be the minimum total length of a forest on $M \cup X$ whose every component meets M. The roots may remain in different components. For formal local families $F = (F_X)_X$, define

$$\|F\|_{\mathcal{B}_M} = \sum_X e^{\kappa \ell_M(X)} \|F_X\|_\infty, \qquad (FG)_Z = \sum_{X \cup X' = Z} F_X G_{X'}. \tag{31}$$

Every F_X is holomorphic in $A|_X$. The product-domain assumption gives its uniform local-function norm. The inequality $\ell_M(X \cup X') \le \ell_M(X) + \ell_M(X')$ makes this a Banach algebra. Enlarging the root set is contractive. Evaluation, obtained by summing the local coefficients, is a bounded algebra homomorphism.

Expand $C = C_0 - C_0 B C_0 + C_0 B C_0 B C_0 - \cdots$ in the individual precision labels B_Y. Retain these labels before grouping by the union of their supports. For endpoints $b, c \in M$, every term connects all its background labels to its endpoints by covariance links and trees inside the supports Y. If their

total length is L_{path}, both $\ell_M(\cup Y) \le L_{\mathrm{path}}$ and $d(b,c) \le L_{\mathrm{path}}$. Successive weighted matrix sums give $K^{j+1}\nu^j$ at order j. Therefore, for $\Delta C = C - C_0$,

$$\max\left\{\sup_{b\in M}\sum_{c\in M} e^{2md(b,c)}\|(\Delta C)_{bc}\|_{\mathcal{B}_M}, \sup_{c\in M}\sum_{b\in M} e^{2md(b,c)}\|(\Delta C)_{bc}\|_{\mathcal{B}_M}\right\} \le v. \tag{32}$$

This bounds the whole covariance marginal, including its background dependence, rather than only visible forest edges.

3.3 Molecule masks and cell endpoint polydiscs

At a molecule BKAR point, its induced cell mask σ is one inside every primitive molecule and has the BKAR values between molecules. Each molecule partition mask is exactly a cell partition mask whose blocks are unions of molecules. Thus σ is in the convex hull of cell partition masks and $0 \le \sigma_{bc} \le 1$.

Choose one cell endpoint pair for each molecule-tree edge, with cell degrees d_b, and use $|z_{bc}| < \delta/[\max(d_b, d_c)J_{bc}]$. The endpoint graph need not span or connect every cell. Let M_0 be the resulting masked and perturbed baseline. The row-sum argument gives

$$\operatorname{Herm} M_0 \ge \lambda I/2, \qquad \|M_0\| \le 2\Lambda_0, \qquad \|M_0\|_{\mathrm{Schur},m} \le K + \delta. \tag{33}$$

For the last bound, each selected edge satisfies

$$e^{md(b,c)}|z_{bc}|\|(C_0)_{bc}\| \le \frac{\delta}{\max(d_b, d_c)} e^{-\kappa d(b,c)}.$$

Its row sum is at most δ, using only the number of incident selected edges. No spanning-tree property on the cell set has been used.

Conjugation by $D_{bb} = e^{sd(b,c)}$ changes M_0 in operator norm by at most $2s(K+\delta)/(em) \le a/2$. Since its Hermitian part is at least a, the inverse entries are bounded by $(2/a)e^{-sd(b,c)}$, and hence

$$\|M_0^{-1}\|_{\mathrm{Schur}} \le L. \tag{34}$$

Apply the same mask and increments to ΔC, obtaining an algebra-valued matrix E. The mask part has norm at most v. By (32), the selected-edge part has row and column sums at most $\delta v/K$:

$$|z_{bc}| v e^{-2md(b,c)} \le \frac{\delta v}{K \max(d_b, d_c)} e^{-md(b,c)}.$$

Consequently

$$\|E\|_{\mathrm{Schur}(\mathcal{B}_M)} \le v_1. \tag{35}$$

These are precisely the estimates needed for the density expansion, and they remain valid for the molecule masks and the original cell endpoint polydiscs.

3.4 The entire density, including determinant normalization

Put $F = M_0^{-1}E$. In the matrix support algebra,

$$(M_0 + E)^{-1} - M_0^{-1} = \sum_{j\ge 1}(-F)^j M_0^{-1},$$

whose Schur norm is at most w, while

$$\|\operatorname{Tr}\log(I+F)\|_{\mathcal{B}_M} \le D|M|[-\log(1-u)].$$

The exact normalized density identity is

$$p_{M_0+E}(x) = p_{M_0}(x) \exp\left[-\tfrac{1}{2}\operatorname{Tr}\log(I+F) - \tfrac{1}{2}x^T\big((M_0+E)^{-1} - M_0^{-1}\big)x\right]. \tag{36}$$

Taking its algebra norm, the determinant factor costs at most $2^{D|M|/2}$. The quadratic factor costs at most $e^{w\|x\|^2/2}$, which by (26) uses at most half of the baseline Gaussian decay. It follows that

$$\|p_{M_0+E}(x)\|_{\mathcal{B}_M} \le \widehat{K}_G^{|M|} p_{2V_* I}(x_M). \tag{37}$$

Here $p_{2V_* I}$ is the independent product Gaussian density with that covariance in every original cell. All inverse, logarithm, exponential and support series are absolutely convergent under the displayed majorants.

For an interaction supported on $Y \subset M$, $\ell_M(Y) = 0$. Thus w_Y is an algebra element with norm at most v_Y. A primitive interaction molecule on S has norm at most M_S from (10); a bad singleton has empty background support. Integrating (37) against their product gives on the entire endpoint polydisc

$$\left\| \mathbb{E}_{M_0+E} \prod_a f_{S_a} \right\|_{\mathcal{B}_M} \le \prod_a \left(\widehat{K}_G^{|S_a|} M_{S_a} + \widehat{\tau} \mathbf{1}_{\{|S_a|=1\}} \right). \tag{38}$$

3.5 Factorization with overlapping background labels

Retain independent formal scalar tags on the precision labels, and on the interaction labels if desired, until absolute convergence is proved. At a forest mask the complete marginal Gaussian matrix is block diagonal over the forest components. Both its determinant and inverse therefore factorize in (36), giving an identity of formal power series. This remains true when different components read overlapping background coordinates. Grouping coefficients by the union of their background labels respects multiplication. The density for a component is its actual marginal density, with its own root set; injection into the algebra with all moment roots is contractive.

Thus forest factorization holds in the support representation itself, not merely after evaluating A. Banach-valued Cauchy applied to (38) yields the same cell-factorial bound (16). The overlap-animal and planted-tree estimates give

$$\sup_i \sum_{M \ni i, X} e^{\kappa \ell(M) + \kappa \ell_M(X) + 2|M|} \|\zeta(M; X)\|_\infty \le \widehat{\eta}. \tag{39}$$

Hard-core compatibility is still disjointness of the marked supports M, and does not require disjoint background supports.

For a connected Mayer cluster, the root and union inequalities give

$$\ell(\cup M_a) \le \sum_a \ell(M_a), \qquad \ell_{\cup M_a}(\cup X_a) \le \sum_a \ell_{M_a}(X_a).$$

The same pinned cluster estimate therefore yields

$$\log Z_V = \sum_{M \neq \varnothing, X} U(M; X)(A|_X), \qquad \sup_i \sum_{M \ni i, X} e^{\kappa[\ell(M) + \ell_M(X)] + |M|} \|U(M; X)\|_\infty \le 2\widehat{\eta}. \tag{40}$$

3.6 Re-rooting at the final connected background support

Choose deterministic minimizing trees for M, minimizing forests attaching X to M, and shortest lattice paths for their edges. Their union $H(M, X)$ is connected and contains $M \cup X$. With $L_* = \ell(M) + \ell_M(X)$,

$$|H(M, X)| \le |M| + L_*, \qquad d(a, b) \le L_* \quad (a \in M,\ b \in H(M, X)).$$

Consequently

$$e^{a_*|H|} \le e^{|M|+\kappa L_*} e^{-(\kappa/2)d(a,b)}. \tag{41}$$

Group $U(M;X)$ by this fixed hull to define $E_{\Lambda,H}$. To root a sum at $b \in H$, overcount by all $a \in M$ and use (40); the lattice convolution costs only $S_{\kappa/2}$. This proves (28).

Under (29), the multiplier interpolation $w_Y \mapsto t w_Y$ is holomorphic in the same support algebra. The Banach-valued Cauchy estimate of the preceding section gives $8\widehat{\epsilon}$ in the marked rooted norm. The fixed hull map has norm at most $S_{\kappa/2}$ by (41), proving (30).

3.7 What the locality theorem does and does not identify

The activities may depend on the whole background-independent finite-volume baseline $C_{0,\Lambda}$. Their variable background dependence reads only $A|_H$, but volume consistency is not claimed. A baseline chosen separately at each arbitrary background would localize only perturbations from that reference. The product background domain and the small local precision norm are substantive hypotheses, not consequences of a weak global gauge-field norm or of covariance decay alone.

A deterministic path choice may preserve translations while breaking discrete rotations; arbitrary decompositions do not make each activity gauge invariant. An application needing these symmetries must supply an equivariant construction or an admissible averaging argument. The periodic extension later in this notebook verifies uniform volume counting and the required path/hull construction with the same length and cardinality bounds, including short tori. No actual regulator coordinates, source coefficients, moving gauge chart or admitted history family has been identified by the abstract hypotheses alone.

4 A tunable density cost near a real Gaussian baseline

The fixed density factors in the preceding theorems can cost $\exp(cD)$ per interaction cell. That cost need not fit an application's support-decay exponent when a cell contains many coordinates. A real baseline allows a tunable factor $\exp(O(\theta D))$. The dominating Gaussian is correlated; a joint-tail estimate replaces independence.

Lemma 4.1 (Relative Gaussian density)**.** *Let C_r be a real symmetric positive covariance in N dimensions, and let*

$$M = C_r^{1/2}(I+F)C_r^{1/2}, \qquad F^T = F, \qquad \|F\| \le \theta < 1.$$

Then

$$|p_M(x)| \le \left(\frac{1+\theta}{1-\theta}\right)^{N/2} p_{(1+\theta)C_r}(x). \tag{42}$$

Proof. The determinant modulus obeys $|\det(I+F)| \ge (1-\theta)^N$. Moreover

$$(1+\theta)\operatorname{Herm}(I+F) - (I+F)^*(I+F) = \theta I - (1-\theta)\operatorname{Herm} F - F^*F \ge 0,$$

using $\operatorname{Herm} F \le \theta I$ and $F^*F \le \theta^2 I$. Congruence by $(I+F)^{-1}$ gives $\operatorname{Herm}(I+F)^{-1} \ge (1+\theta)^{-1} I$. Insert these two bounds in the Gaussian density, including the normalization of the real density on the right. □

Lemma 4.2 (Correlated block tails)**.** *Let G be a centered real Gaussian on D-dimensional cells, with covariance at most $\Lambda_{\mathrm{g}} I$. For $0 < \sigma \le 1$ set*

$$p_{\sigma,R}(\Lambda_{\mathrm{g}}) = (1+\sigma)^{D/2} \exp\left[-\frac{\sigma R^2}{2(1+\sigma)\Lambda_{\mathrm{g}}}\right]. \tag{43}$$

For every finite cell set B,

$$\mathbb{P}\{\|G_i\|_2 > R \text{ for all } i \in B\} \le p_{\sigma,R}(\Lambda_{\mathrm{g}})^{|B|}. \tag{44}$$

Consequently, for $a_i \ge 0$,

$$\mathbb{E}\prod_{i\in I}(a_i + \mathbf{1}_{\{\|G_i\|_2>R\}}) \le \prod_{i\in I}(a_i + p_{\sigma,R}(\Lambda_{\mathrm{g}})). \tag{45}$$

Proof. On the event in (44), $\sum_{i\in B}\|G_i\|^2 > |B|R^2$. Exponential Markov with $t = \sigma/[2(1+\sigma)\Lambda_{\mathrm{g}}]$ gives

$$\mathbb{P}(\text{event}) \le e^{-t|B|R^2}\det(I - 2tC_B)^{-1/2} \le e^{-t|B|R^2}(1+\sigma)^{D|B|/2}.$$

Expand the product into choices of bad singleton cells to obtain (45). No independence is assumed. □

4.1 The marked theorem for a small complex covariance family

Let C_r be a fixed background-independent real covariance with $\lambda I \le C_r \le \Lambda_0 I$. Suppose the actual holomorphic complex symmetric covariance satisfies

$$\sup_A \|C(A) - C_r\| \le \theta\lambda/2, \qquad 0 < \theta \le 1/4. \tag{46}$$

Retain the actual covariance majorant J and A_κ from (2), the fixed cutoff, and the bounded local interaction multipliers of the marked theorem. For any fixed $0 < \sigma \le 1$ replace its constants by

$$\begin{aligned}
&\delta_\theta = \theta\lambda/2, && K_\theta = \left(\frac{1+\theta}{1-\theta}\right)^{D/2} = e^{D\,\mathrm{artanh}\,\theta}, && \\
&p_{\theta,\sigma,R} = p_{\sigma,R}((1+\theta)\Lambda_0), && \tau_\theta = K_\theta p_{\theta,\sigma,R}, && c_\theta = e^2/\delta_\theta, \\
&\epsilon_\theta = \sup_i \sum_{Y\ni i} v_Y K_\theta^{|Y|} e^{4|Y|+\kappa\ell(Y)}, && \eta_\theta = e^3\tau_\theta + \epsilon_\theta, && \eta_{\theta,*} = \frac{1}{4\max(1, c_\theta A_\kappa)}.
\end{aligned} \tag{47}$$

Theorem 4.3 (Tunable marked expansion)**.** *If $\eta_\theta \le \eta_{\theta,*}$, the conclusion (6) holds with $2\eta_\theta$. Under the quarter budgets $e^3\tau_\theta \le \eta_{\theta,*}/4$ and $\epsilon_\theta \le \eta_{\theta,*}/4$, the interaction-difference bound is $8\epsilon_\theta$.*

Proof. A cell partition mask is a direct-sum pinching, and therefore contracts the operator norm of $C(A) - C_r$. Convex combinations of these masks have the same property. At a molecule BKAR mask the covariance differs from its real masked baseline by at most $\theta\lambda/2$. The degree-adapted cell endpoint polydisc, with radius budget δ_θ and the actual majorant J, adds at most another $\theta\lambda/2$. The total relative perturbation from the real masked baseline has norm at most θ.

Lemma 4.1 applies. Its dominating real Gaussian covariance is at most $(1+\theta)\Lambda_0 I$. Bound nonsingleton molecule functions by M_S, and expand the singleton bounds $M_i + \mathbf{1}_{\mathrm{bad},i}$. Lemma 4.2 gives exactly the product bound used in the cell-endpoint Cauchy estimate, with

$$b_\theta(S) = K_\theta^{|S|}M_S + \tau_\theta\mathbf{1}_{\{|S|=1\}}.$$

The overlap-animal, planted-tree, Mayer and interpolation arguments are unchanged. □

If $C(A)$ is identically the real baseline, one may use the larger budget $\delta_\theta = \theta\lambda$, because no background deviation consumes half of it. Operator closeness suffices for this marked theorem but does not imply strict background locality.

4.2 The tunable strictly local version

Assume the local precision and product-domain hypotheses of Section 3, now with C_0 real symmetric positive. Retain K, ν, v, $J_{bc} = Ke^{-md(b,c)}$ and A_κ. Fix $0 < \theta \le 1/4$ and replace the precision constants by

$$\begin{aligned} &\delta = \theta\lambda, && \bar{K} = K + \delta, && a = \lambda/2, \\ &s = \min\{m/2,\ aem/(4\bar{K})\}, && L = \frac{4}{\lambda}\sum_z e^{-s|z|}, && \\ &v_1 = v(1 + \delta/K), && u = Lv_1, && w = \frac{L^2 v_1}{1-u}. \end{aligned} \tag{48}$$

Require $K\nu < 1$ and the strengthened smallness conditions

$$u \le \theta, \qquad w \le \frac{\theta}{(1+\theta)\Lambda_0}. \tag{49}$$

The earlier condition $v_1 \le \lambda/4$ follows because $L \ge 4/\lambda$ and $\theta \le 1/4$. Define

$$\rho_\theta = \frac{1+\theta}{1-\theta}, \qquad \mathcal{K}_\theta = \left(\frac{1+\theta}{(1-\theta)^3}\right)^{D/2}, \qquad \sqrt{}^{\theta,\sigma,R} = p_{\sigma,R}(\rho_\theta \Lambda_0). \tag{50}$$

In (27) replace $\widehat{K}_G, \widehat{p}_R$ by $\mathcal{K}_\theta$, $\sqrt{}^{\theta,\sigma,R}$, and use $c_G = e^2/\delta$ with the present δ. Denote the resulting quantities by $\tau_\theta, \epsilon_\theta, \eta_\theta, \eta_{\theta,*}$.

Theorem 4.4 (Tunable strictly local expansion)**.** *Under* (49) *and* $\eta_\theta \le \eta_{\theta,*}$, (28) *holds with bound* $2S_{\kappa/2}\eta_\theta$. *Under the corresponding quarter budgets, its interaction difference is bounded by* $8S_{\kappa/2}\epsilon_\theta$. *All domain, volume and symmetry scope statements of Section 3 remain in force.*

Proof. Let $C_{r,M}$ be the real masked baseline on a moment set M. It is between λI and $\Lambda_0 I$. Add only its selected complex cell endpoint perturbations to form M_0. Their norm is at most $\theta\lambda$, so

$$|p_{M_0}(x)| \le K_\theta^{|M|} p_{(1+\theta)C_{r,M}}(x).$$

The support-algebra bounds on the source-dependent matrix E from Section 3 are unchanged with the present constants. In particular the determinant correction has norm at most $(1-u)^{-D|M|/2}$ and the inverse correction costs at most $\exp(w\|x\|^2/2)$.

Since $C_{r,M} \le \Lambda_0 I$, $\|x\|^2 \le \Lambda_0 x^T C_{r,M}^{-1} x$. Condition (49) therefore gives

$$p_{(1+\theta)C_{r,M}}(x) e^{w\|x\|^2/2} \le \left(\frac{\rho_\theta}{1+\theta}\right)^{D|M|/2} p_{\rho_\theta C_{r,M}}(x).$$

Combining this inequality with $u \le \theta$ and the baseline density factor proves

$$\|p_{M_0+E}(x)\|_{\mathcal{B}_M} \le \mathcal{K}_\theta^{|M|} p_{\rho_\theta C_{r,M}}(x). \tag{51}$$

The dominating Gaussian is real and correlated, with covariance at most $\rho_\theta \Lambda_0 I$. Expand the bad singleton choices and use Lemma 4.2. This gives (38) with the new constants. All cell-endpoint, formal-factorization, Mayer, hull and bounded-multiplier interpolation arguments then apply. □

4.3 The dimension tradeoff

For $\theta \le 1/4$,

$$\log K_\theta \le \frac{16}{15} D\theta, \qquad \log \mathcal{K}_\theta = \frac{D}{2}[\log(1+\theta) - 3\log(1-\theta)] \le \frac{5}{2} D\theta.$$

Thus $\theta = 1/(4D)$ bounds these unmarked per-cell factors by $e^{4/15}$ and $e^{5/8}$, independently of D. The covariance-edge constant grows like θ^{-1}, and the local precision neighborhood must meet (49). If the other baseline constants stay fixed, this exchanges an exponential-in-D interaction support cost for a polynomial edge-budget loss and a smaller precision neighborhood.

The choice $\sigma = 1$ retains the faster Gaussian tail $2^{D/2} e^{-R^2/(4\Lambda_{\mathrm{g}})}$. Alternatively, $\sigma = 1/(4D)$ bounds its prefactor by $e^{1/8}$, but its exponent is only of order $R^2/(D\Lambda_{\mathrm{g}})$. The cutoff must satisfy the resulting tail budget in either case. The lemma does not show that the actual source covariance, coordinate family, local precision or cutoff admits these tuned neighborhoods. It removes a specific artificial density cost while keeping those hypotheses explicit.

5 Periodic volumes and short periods

The preceding strict-support expansions also hold on finite rectangular periodic lattices. This is a geometric extension of the abstract theorems; their covariance, precision, interaction and background-domain assumptions must still hold for the actual periodic data.

Proposition 5.1 (Periodic strict-support expansion)**.** *Let*

$$\mathbb{T}_N = \prod_{\mu=1}^{d} \mathbb{Z}/N_\mu\mathbb{Z}, \qquad N_\mu \ge 1,$$

with its nearest-neighbor graph distance d_N. One may replace the finite subset of $\mathbb{Z}^d$ in the strictly local overlap theorem and its tunable real-baseline version by $\mathbb{T}_N$, or by any subset of it. Use d_N in all weighted matrix norms and in the metric tree and rooted forest lengths. If the displayed hypotheses hold with constants independent of N, both conclusions hold with exactly the same constants. The resulting supports are connected subsets of $\mathbb{T}_N$, including when some periods equal one or two.

Proof. For every base point $b \in \mathbb{T}_N$, choose one shortest integer representative $r_b(c) \in \mathbb{Z}^d$ of each difference $c - b$. The choices form an injection and satisfy $|r_b(c)|_1 = d_N(b,c)$. Consequently, for every nonnegative function f on the nonnegative integers,

$$\sum_{c \in \mathbb{T}_N} f(d_N(b,c)) \le \sum_{z \in \mathbb{Z}^d} f(|z|_1). \tag{52}$$

For sums excluding $c = b$, the zero representative is excluded on the right. In particular, all exponential lattice-sum bounds used for A_κ, the inverse Schur constant, and $S_{\kappa/2}$ retain their previous upper bounds.

The weighted matrix product estimates require only the triangle inequality. The inverse Schur proof uses the weight $D_{bb} = \exp(s d_N(b,c))$ and the inequality

$$|d_N(b,c) - d_N(b',c)| \le d_N(b,b'). \tag{53}$$

Thus its conjugation estimate, operator inverse bound and entry decay are unchanged; summing entries uses (52). Partition masks, cell endpoint degrees, Gaussian density comparisons, the rooted support algebra, and the overlap and Mayer combinatorics are independent of any choice of integer representatives. They therefore apply directly on the finite quotient.

It remains to check the conversion to connected background support. For each nonempty marked set M and background support X, let $\ell_N(M)$ be its minimum metric tree length, and let $\ell_{N,M}(X)$ be the minimum forest length on $M \cup X$ with every component meeting M. Realize fixed minimizing trees and forests by fixed shortest torus paths, and let $H(M,X)$ be their union. It is a connected subset of $\mathbb{T}_N$, contains $M \cup X$, and, writing $L_* = \ell_N(M) + \ell_{N,M}(X)$, satisfies

$$|H(M,X)| \le |M| + L_*, \qquad d_N(a,b) \le L_* \quad (a \in M,\ b \in H(M,X)). \tag{54}$$

The vertex bound follows by adding the selected paths one edge at a time; identifications caused by wrapping can only reduce the number of vertices. If the original cell set is a proper subset of the torus, the hull may leave that set, while the coefficient continues to read only the original background coordinates in X.

For $a_* = \min(1/2, \kappa/4)$, (54) gives

$$e^{a_*|H(M,X)|} \le e^{|M|+\kappa L_*} e^{-(\kappa/2)d_N(a,b)}. \tag{55}$$

Overcounting by the marked anchors $a \in M$ and applying (52) costs at most the same $S_{\kappa/2}$ as before. This proves both the connected-support norm and the interaction-difference bound with the original constants. □

The proof does not use a positive injectivity radius or distinct positive and negative neighbors. Repeated interaction labels whose supports become equal after a periodic identification are still counted as distinct labels, as in the original multiplier norm. The theorem assumes the corresponding actual periodic coefficient norms; it does not infer them merely by relabeling an infinite-volume model. Deterministic shortest-path tie choices need not preserve translations or rotations, and no such activity symmetry is asserted here. Nor does this extension assert consistency between different periodic volumes.

6 Using the published localization and integration theorem

There is an existing specialized integration route in the literature. Bałaban's *Renormalization group approach to lattice gauge field theories II*, Commun. Math. Phys. **116** (1988), Lemma 2, pp. 10–11, equations (1.41)–(1.43), localizes the complete fluctuation exponent, including the difference of inherited effective interactions. Lemma 3, p. 20, equation (2.38), and the logarithmic estimate (2.41), p. 21, integrate these overlapping terms. These results can be used in their published setting; another abstract integration theorem is not a prerequisite for using them.

6.1 The actual coefficient and field domains

The original domain is equation (1.34), p. 9:

$$\mathcal{U}^c_{k+1}(Y, (1+\beta)\alpha_0, (1+\beta)\alpha_1, \alpha_0) \ \times\ \{B : |B| < \epsilon_1/g \text{ on } Y\}.$$

The first factor is an orbit domain from CMP109, pp. 262–263, equations (1.10)–(1.16). Its restrictions include reconstructed backgrounds on subregions. It cannot be replaced solely by an unverified global supremum condition. The operator and minimizer walk deformations in CMP116, pp. 3–6, are also inputs.

To distinguish two scales, let M_{loc} denote the fixed localization cube size in that paper; it is different from the refinement $M = L^k$ in our flat covariance theorems. The local exponent is organized as

$$P^{(k)} + \text{inherited-action difference} = \sum_Y V_Y, \qquad V_Y = \tfrac{1}{2}\langle B, Q_Y(U,J,B)B\rangle + V''_Y(U,J,B).$$

The displayed source bounds are (1.43) for the quadratic kernel and (1.36) for the remainder. Their localization weight is a normalized minimum tree length meeting the M_{loc} cubes of Y. It is stronger than a diameter bound and comes with the source's specific geometric summations. The source exponent contains the Haar, constraint Jacobian and gauge contributions; it is not only the bare Wilson cubic remainder.

The original article was inspected through its extracted text. Some displayed numerical coefficients are lost in that extraction; the numbered source bounds are therefore located here rather than reconstructed by guesswork. The matching deposited formulas are Raw I:30590–30597 and Raw II:3190–3208. The latter attributes (1.43) to CMP109, but that formula in this argument is CMP116 (1.43), as also recorded at Raw II:11178–11188.

6.2 The logarithmic-radius transfer

Fix all geometric and analytic parameters before choosing the coupling ceiling. For $R(g) = c(\log g^{-2})^p$, $p > 1/2$,

$$gR(g) < \epsilon_1/2, \qquad gR(g)^3 \longrightarrow 0, \qquad e^{-c_1 R(g)^2} = o(g^q)$$

for every fixed $q > 0$. Thus cutoff points in the actual source variable lie inside its fixed physical-field analytic radius. For localized coefficients holomorphic on that radius and vanishing at the indicated orders, radial Schwarz estimates give

$$\|Q_Y(gB)\| \le a_Y gR, \qquad |V_Y''(gB)| \le a'_Y gR.$$

Fixed inverse powers of the analytic radius are included in a_Y, a'_Y. On the cutoff this bounds V_Y by gR^3 times a local coefficient and a polynomial support factor. A reserved part of the tree-decay exponent pays for that polynomial. Raw II:11178–11195 uses this distinction between cubic and merely vanishing contributions.

Changing the real cutoff nevertheless changes the integral. In the original integration proof the Gaussian bad-bond gain must be rechecked at the new threshold in (2.22), (2.26) and (2.31). Its numerical scale is $e^{-cR(g)^2}$, which pays all fixed geometric entropy below a sufficiently small coupling ceiling. This numerical fact alone does not verify preservation of the source's localized operator representations. The boundary-specific domain CMP119 (3.43) is not a replacement for the whole-lattice interior domain (1.34).

6.3 Why the tunable density estimate matters

The first overlap theorem above has a sufficient interaction norm containing $K_G^{|Y|}$. If a source localization cube is one cell, its real coordinate dimension is $D = O(M_{\mathrm{loc}}^4)$, and the original $\log K_G$ has that dimension cost. Taking current-unit cells instead produces the same loss through their number in Y. The published coefficient decay does not automatically pay this fixed cost: its parameter hierarchy may choose M_{loc} large after the decay exponent has been fixed. Decreasing the coupling cannot cure an unfavorable exponential support exponent.

The tunable real-baseline comparison proved in this notebook reduces that cost to $O(\theta D)$ per source cell. Choose θ of order D^{-1}, with the source cubes as cells. Its Cauchy derivative constant then grows with θ^{-1}, but is fixed before the coupling is decreased. This removes the automatic M_{loc}^4 loss in the support exponent. The strict version also requires the displayed precision perturbation smallness at that chosen θ.

For example, if a source coefficient bound has sufficient spare tree decay to pay the resulting fixed support weight and its geometric summation, the Schwarz estimate above implies $\epsilon_\theta(g) \le C_\theta gR(g)^3$. The interaction and Gaussian-tail smallness conditions then hold together below a fixed coupling ceiling.

This is a precise conversion criterion; the required source decay and precision domain must still be matched. Alternatively one can retain CMP116's own integration theorem, whose controllably small quadratic and determinant errors serve the same purpose.

6.4 The immediate model-specific proof

The flat current-scale covariance is supplied by Notebook 17. Notebook 19 constructs its nonflat physical-background version, the exact relative-field pivot chart, and the local measure factors on the homogeneous small-global-potential branch. Notebook 20 verifies the full local coefficient and precision match for the bare first-step Wilson small-field integral. To invoke the original CMP116 theorem on its larger domains, one must still show that the background, independent current and every restricted reconstruction lie in its orbit domain, and match the source's deformed operators. If a cutoff or inherited carrier is changed, its exact transfer through that proof must be established.

The retained cutoff coordinates in Notebook 19 are preserved exactly. Additional indicators retain their own pullbacks. The direct first-step proof does not establish the independent current and mixed-history domains, transport of support to the next coarse background, or the later quantum interactions. Those are further model statements needed to close the induction.

7 The model estimate to close next

For a local analytic physical-field remainder $F_Y(z)$ vanishing through degree two, bounded by a_Y for $\|z\|_{\infty,Y} < r_0$, the one-variable Schwarz lemma on each complex radial line gives

$$|g^{-2}F_Y(gx)| \le a_Y r_0^{-3} g R^3 \qquad (\|x\|_{\infty,Y} \le R).$$

If the corresponding weighted local sum of a_Y is finite, the interaction parameter in this notebook tends to zero with gR^3. A merely extensive bound on the complete analytic action does not establish that local sum. The full fluctuation exponent also contains Jacobian terms beginning at first order and Haar terms beginning at second order; their cutoff bounds must be included with their actual prefactors.

Notebook 19 supplies the physical homogeneous Gaussian and its local measure and gauge factors. Notebook 20 verifies the full coefficient and precision hypotheses for the bare first Wilson small-field integral. Notebook 21 supplies the localized classical action and precision uniformly in earlier refinement and closes one classical-reference integration. The remaining model work is to control the inherited quantum interactions, transport the activities to the next coarse background, handle the complement of the small-field domain, and close the renormalization induction. The continuum existence argument requires that induction together with the observable and reconstruction obligations of Notebook 01.

References

[1] A. Abdesselam and V. Rivasseau, *Trees, forests and jungles: a botanical garden for cluster expansions*, arXiv:hep-th/9409094.

[2] A. Abdesselam, *Notes on the Brydges–Kennedy–Abdesselam–Rivasseau forest interpolation formula*, author notes, Theorem 1 and the convex hull of partition masks.

[3] R. Kotecký and D. Preiss, *Cluster expansion for abstract polymer models*, Commun. Math. Phys. **103** (1986), 491–498, doi:10.1007/BF01211762.

[4] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 3–7 and 17 in this workspace. The Gaussian cutoff and strict background-locality constructions are separate inputs to the present extension.

[5] T. Bałaban, *Renormalization group approach to lattice gauge field theories. II. Cluster expansions*, Commun. Math. Phys. **116** (1988), 1–22, original article copy.

[6] T. Bałaban, *Renormalization group approach to lattice gauge field theories I*, Commun. Math. Phys. **109** (1987), 249–301, original article copy.